THE PRACTICAL MONEY SYSTEM

A Beginner's Guide to Budgeting, Saving, Debt Payoff & Building Wealth



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MONEY SYSTEM

A Beginner's Guide to Budgeting, Saving, Debt Payoff & Building Wealth

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This book is general financial education. It is not personalised financial, tax, legal or investment advice, and it does not recommend any product. Investing involves risk, including loss. Rules and taxes differ by country. Examples use assumed figures.

Sources

Statements of fact are supported by the references at the end of the book, with the date each source was checked. Web pages change; confirm current details before relying on them.

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Introduction

Most money guides teach one skill and leave you to guess how it connects to the rest. You learn to budget, but not what to do with the surplus. You learn to invest, but not whether to do it while carrying a credit card balance. This book puts the whole sequence in one place and in order: understand your cash flow, build a budget that survives real life, set aside an emergency fund, deal with debt, create saving systems, understand compound interest, and only then learn the basics of investing, risk, scams and long-term planning.

It is written for people who are starting out or starting over: students, young professionals, freelancers with uneven income, and anyone who never had these things explained clearly. It works in any currency. Every example uses the neutral unit **CU** (currency unit), so 1,000 CU can be dollars, euros, rupees or pounds. Swap in your own numbers.

Important: education, not advice

This book is general education. It is not personalised financial, tax, legal or investment advice, and the author does not know your circumstances. It does not recommend any investment, fund, broker, bank, app or product, and it does not tell you what to buy or sell. Nothing here guarantees any result. Investments can lose value, and past results do not predict future results. Every interest rate, return and price in this book is an **assumption chosen for illustration**, not a forecast and not a current market figure.

Rules differ by country: tax-advantaged accounts, deposit protection, consumer credit law, debt relief processes, insurance and inheritance rules are all local. Where this book gives a country example, it is labelled and sourced. For everything else, check your own country's financial regulator or consumer protection agency, and consider a qualified adviser before making a large decision.

How the book is organised

Chapters 1 to 3 give you the foundation: a clear view of your money and a budget. Chapters 4 to 7 cover protection and saving: the emergency fund, debt, repayment strategies and saving systems. Chapters 8 to 10 explain compound interest, investing and risk. Chapter 11 covers scams, and Chapter 12 covers long-term planning. The Conclusion gives you a 90-day plan, and the appendices hold the worksheets and glossary in copyable form.

How to use it

Read one chapter, then do its exercise before moving on. Worksheets are in fenced text blocks so you can copy them into a spreadsheet, a notebook or a notes app. The worked examples show every step so you can check the arithmetic yourself, and you can repeat them with your own figures.

Rounding convention: money is rounded to two decimals (half up); where interest is added monthly, it is rounded to the cent each month. Percentages are shown to two decimals when precision matters.

What this book does not do

It does not pick investments, promise returns, offer credit repair, or replace a debt adviser, accountant, lawyer or financial planner. It does not assume you live in a particular country.

How this book was researched

On 2026-11-07 the book was planned against a review of listings for competing budgeting, debt and money guides, to see what they leave out (most are single-topic workbooks or short guides), and official consumer-protection pages were read for the emergency fund, debt and scam chapters (listed in Appendix D). Every calculation was produced and checked with a Python script (`verify_math.py`), and each worked example matches the script's output exactly.

CHAPTER 1

Understanding personal money management

What a money system is

A money system is a small set of repeating actions that decide where each unit of income goes before you are tempted to spend it. It has four jobs:

1. **See:** know what comes in, what goes out and what you owe.
2. **Plan:** give every unit of income a job (a budget).
3. **Protect:** hold a cushion so a bad month does not become a debt.
4. **Grow:** reduce costly debt, then save and invest for later goals.

The order matters. Investing before you have a cushion often means selling at a bad time to cover an emergency. Saving aggressively while paying 25 percent interest on a card usually loses ground. The sequence in this book is a common logical order, not a law. Your situation may justify changing it.

The five numbers that matter

You can run a sound system on five figures. Calculate them in Chapter 2 and update them monthly.

- **Net income:** money that actually arrives after tax and deductions.

- **Essential spending:** what you must pay to live and work (housing, food, transport, utilities, insurance, minimum debt payments).
- **Savings rate:** savings divided by net income.
- **Debt balance and its interest cost:** what you owe and what it costs you per month.
- **Net worth:** assets minus liabilities.

Cash flow versus wealth

Cash flow is money moving through a period (a month). Wealth is what is left standing at a point in time (net worth). People confuse the two. A high income with high spending creates a lot of cash flow and little wealth. A modest income with a steady surplus builds wealth slowly. The rest of the book is about turning cash flow into protection, then into net worth.

Needs, wants and the grey zone

Needs are costs you cannot reasonably avoid; wants are optional. Many items are in between: a phone plan is a need, the premium tier is a want. A useful habit is to split each category into a floor (the cheapest acceptable version) and an upgrade. Budget the floor as essential and the upgrade as personal spending.

Money and behaviour

Most budget failures are not arithmetic failures. Common causes:

- **Plans made for a perfect month.** The budget has no room for irregular costs.
- **Invisible spending.** Small repeated purchases and subscriptions are not tracked.
- **All-or-nothing rules.** One overspend leads to abandoning the system.
- **Social spending pressure.** Costs set by friends, family or status.

The fixes are practical: include a buffer line, track for a month before judging, expect a bad week and adjust rather than quit, and decide in advance how much you will spend on social life.

Exercise 1: Write your starting statement

```
MY STARTING STATEMENT (date: _________)
1. Money I received last month (after tax): _______ CU
2. Money I spent last month (best estimate): _______ CU
3. Debts I owe (total): _______ CU
4. Cash and savings I hold: _______ CU
5. What worries me most about money: ____________________
6. One money goal for the next 90 days: ____________________
```

Do not try to be exact yet. The aim is an honest starting line.

CHAPTER 2

Income and expenses

Step one: list your income

Write every source of money, in the form it arrives (net).

- **Salary or wages:** use take-home pay, not the headline figure.
- **Self-employment or freelance income:** use what you receive after you set aside tax (see below).
- **Other income:** allowances, side work, benefits, rental income, interest.

If income varies, do not use your best month. Use the method for irregular income later in this chapter.

Step two: list your expenses

Group expenses by how they behave.

Fixed expenses stay about the same each month: rent or mortgage payment, insurance premiums, loan minimums, subscriptions, phone contract.

Variable expenses change: groceries, transport, utilities, eating out, clothes, entertainment.

Irregular expenses are real but do not arrive monthly: annual insurance, car repairs, medical costs, gifts, travel, replacement of devices. These are the main reason budgets fail, because a monthly

view hides them. Chapter 7 covers sinking funds, which turn them into small monthly amounts.

Cash flow in one line

```
Monthly cash flow = net income received - total spending
```

Positive means you have a surplus to allocate. Negative means you are funding spending with savings or debt, which is the first thing to fix.

Tracking: three methods

1. **Bank statement review.** Download the last two or three months of statements and categorise each line. Fastest way to start.
2. **Running log.** Record each purchase when it happens, in a notes app or spreadsheet.
3. **Envelope or cash method.** Withdraw cash for variable categories and spend only that. Works well for people who overspend on cards, but is less practical where cash is rarely used.

Pick one and run it for at least 30 days before you decide how to budget.

Building the tracker in a spreadsheet

Use a sheet with these columns:

```
Date | Description | Category | Amount | Type | Payment method
```

Rules for building it:

- Enter income as positive and spending as negative, or use the Type column (Income, Fixed, Variable, Irregular). Be consistent.

- Keep a short list of categories (10 to 12 is enough) in a separate sheet and use a dropdown for the Category column so spellings stay identical.
- Create a summary sheet with a pivot table or SUMIF formulas that total each category by month, for example `=SUMIFS(Amount, Category, "Groceries", Month, "2026-11")`.
- Add a Month column computed from the date so you can group by month.
- Review weekly for 10 minutes. A tracker you never open is a hobby.

Income and expense tracker layout

Copy this table into your spreadsheet as the summary view.

```
Category        | Type     | Planned | Actual | Difference
----------------|----------|---------|--------|-----------
Income (net)    | Income   |         |        |
Housing         | Fixed    |         |        |
Utilities       | Fixed    |         |        |
Groceries       | Variable |         |        |
Transport       | Variable |         |        |
Insurance       | Fixed    |         |        |
Debt minimums   | Fixed    |         |        |
Eating out/fun  | Variable |         |        |
Subscriptions   | Fixed    |         |        |
Irregular fund  | Irregular|         |        |
Savings         | Savings  |         |        |
TOTAL SPENDING  |          |         |        |
LEFT OVER       |          |         |        |
```

Irregular income for freelancers and gig workers

If income swings month to month, a normal monthly budget fails in the low months. Use this method instead.

1. **Separate tax first.** Move a fixed percentage of every payment into a tax account the day it arrives. The right percentage depends on your country and situation; ask your tax authority or an accountant. The worked example in Chapter 3 uses 20 percent as an **assumption only**.
2. **Find your baseline.** List the last 12 months of net income. Take the average of the lowest six months after tax set-aside, rounded down. This is the cautious baseline: budget on it, not on the average of all months. You may choose a somewhat higher salary only if a buffer in the holding account covers the low months, as Chapter 3 shows.
3. **Pay yourself a salary.** Keep income in a holding account and transfer a fixed amount to your everyday account on the same date each month.
4. **Fill the buffer first.** In good months the surplus stays in the holding account until it covers about one month of your salary, then moves on to the emergency fund and goals.
5. **Review quarterly,** and raise your salary only after the holding account has stayed above the buffer for several months.

Spending analysis exercise

After a month of tracking, calculate each category's share of total spending. Here is an illustrative example with total spending of 2,600 CU:

Category	Amount (CU)	Share of total
Housing	1,100	42.31%
Food at home	420	16.15%
Eating out	230	8.85%
Transport	260	10.00%
Subscriptions	64	2.46%
Other	526	20.23%

Category	Amount (CU)	Share of total
Total	**2,600**	**100.00%**

Calculation: each share is the category amount divided by 2,600, times 100 (for example 230 / 2,600 = 8.85%). The shares add to 100.00%.

Now answer these for your own figures:

```
SPENDING ANALYSIS
1. Which three categories are the biggest? _____________
2. Which category surprised me (more than I guessed)?
_____________
3. Which spending gave me the most value? _____________
4. Which spending gave me the least value? _____________
5. Which single category can I cut by 10 percent without pain?
_______
6. What is 10 percent of that category in CU? _______
7. Which subscriptions have I not used in 60 days? _______
```

Look at the "Other" line in the example: 20.23 percent of spending that is uncategorised. If a catch-all category exceeds 10 percent, split it, because you cannot manage what has no name.

Common mistakes and fixes

Mistake	Fix
Using gross income	Use take-home pay only
Ignoring irregular costs	List them for the year, divide by 12
Too many categories	Cap at 10 to 12
Tracking for one week then judging	Track at least 30 days
Budgeting freelance income on the best month	Use the lowest-six-month average

CHAPTER 3

Building a realistic budget

The method: zero-based with a buffer

A zero-based budget assigns every unit of net income to a line, so that income minus all assigned amounts equals zero. Savings and debt payments are lines too. This is not a promise to spend everything: "assigned to savings" counts.

Steps:

1. Write your net monthly income (the baseline, if it varies).
2. List fixed costs first (housing, utilities, insurance, debt minimums).
3. Add variable essentials (groceries, transport) using your tracked average.
4. Add annual and irregular costs divided by 12.
5. Add savings: emergency fund, then goals.
6. Add personal spending (fun, eating out).
7. Add a buffer line for small overruns (the budget should have one).
8. Check that the total equals income. Adjust personal spending first, then variable essentials. Do not cut the emergency fund to zero.

Percentage rules of thumb (such as splitting income into needs, wants and savings) are only starting points. Use them to sense-check, not to override your actual costs, which vary with city, family size and income.

Worked example A: low income

Assumptions (fictional): net income 1,800 CU per month, single person, shared housing.

Line	CU per month
Housing	620
Utilities and phone	140
Food (groceries)	330
Transport	130
Insurance and health	60
Minimum debt payments	90
Personal and fun	100
Emergency fund	80
Other goals	50
Buffer	200
Total	**1,800**

Check: 620 + 140 + 330 + 130 + 60 + 90 + 100 + 80 + 50 + 200 = 1,800, so income minus total = 0. Fixed costs (housing, utilities, insurance, minimums) are 620 + 140 + 60 + 90 = 910 CU, which is 50.56% of income. That is high, which is why the buffer is large: there is little room to absorb a surprise. Whatever is left in the buffer at month end moves to the emergency fund.

Worked example B: middle income

Assumptions (fictional): net income 3,600 CU per month, one adult.

Line	CU per month
Housing	1,100
Utilities and phone	220

Line	CU per month
Food (groceries)	480
Transport	260
Insurance and health	150
Minimum debt payments	250
Personal and fun	300
Emergency fund	300
Other goals	250
Buffer	290
Total	**3,600**

Check: 1,100 + 220 + 480 + 260 + 150 + 250 + 300 + 300 + 250 + 290 = 3,600. Fixed costs: 1,100 + 220 + 150 + 250 = 1,720 CU, which is 47.78% of income. Savings and goals (emergency fund plus other goals) are 550 CU, or 15.28% of income (550 / 3,600).

Worked example C: irregular freelance income

Assumptions (fictional): twelve months of gross receipts, in CU: 2,400; 3,900; 1,700; 3,100; 2,600; 4,200; 1,500; 3,300; 2,900; 3,600; 2,000; 3,800. A tax set-aside of 20 percent is assumed (illustration only; your real rate depends on your country and situation).

- Total received: 35,000 CU. Average month: 2,916.67 CU.
- Tax set aside (20%): 7,000.00 CU. Left after tax: 28,000.00 CU, average 2,333.33 CU per month.
- Lowest six months, sorted: 1,500; 1,700; 2,000; 2,400; 2,600; 2,900. Average = 13,100 / 6 = 2,183.33 CU gross.
- The cautious baseline is the lowest-six average after the 20% set-aside: 13,100 x 0.80 / 6 = 1,746.67 CU, which rounds down to 1,700 CU. This example deliberately chooses a higher salary of **2,000 CU per month**, between the cautious baseline

(1,746.67) and the all-month after-tax average (2,333.33). That choice is only safe because a buffer exists, as the test below shows. A reader with no buffer should start at 1,700 CU and raise it later.

Budget on the 2,000 CU salary:

Line	CU per month
Housing	800
Utilities and phone	150
Food (groceries)	350
Transport	120
Insurance and health	110
Minimum debt payments	100
Personal and fun	120
Emergency fund	180
Other goals	70
Total	**2,000**

Check: the lines sum to 2,000. Over the year the salary uses 24,000 CU of the 28,000 CU left after tax, leaving a surplus of 4,000 CU in the holding account.

Does the holding account survive the low months? Month by month, after-tax income (80% of receipts) minus the 2,000 CU salary gives a running balance of: -80; 1,040; 400; 880; 960; 2,320; 1,520; 2,160; 2,480; 3,360; 2,960; 4,000. The lowest point is -80 CU in month 1, so a starting buffer of at least 80 CU is needed; a full month of salary (2,000 CU) is a much safer starting buffer because real months do not arrive in a convenient order. This is the reason step 4 above exists.

Monthly budget worksheet

```
MONTHLY BUDGET WORKSHEET        Month: _________
NET INCOME (baseline)                          _______
FIXED
  Housing                                      _______
  Utilities and phone                          _______
  Insurance                                    _______
  Debt minimums                                _______
VARIABLE ESSENTIALS
  Groceries                                    _______
  Transport                                    _______
  Health and personal care                     _______
IRREGULAR (annual total / 12)
  Sinking funds (Chapter 7)                    _______
SAVING
  Emergency fund                               _______
  Goals                                        _______
  Extra debt payment                           _______
PERSONAL
  Fun and eating out                           _______
  Subscriptions                                _______
BUFFER                                         _______
TOTAL ASSIGNED                                 _______
INCOME - TOTAL ASSIGNED (target 0)             _______
```

When the budget does not balance

If expenses exceed income, there are only three levers: raise income, cut costs, or change a big fixed cost (housing, transport, a loan). Cutting small things while ignoring housing or a car loan rarely closes a large gap. Work through your largest three categories first.

Common mistakes and fixes

- **No buffer line:** add one; unused buffer rolls to the emergency fund.
- **Budget written once and never reviewed:** hold a 15-minute review at month end (checklist in Appendix A).

- **Guilt after overspending:** move the money between lines and continue. A budget is a plan you edit.

CHAPTER 4

Emergency funds

What it is for

An emergency fund is money held for unplanned essential costs: income loss, urgent repairs, medical bills, travel for family emergencies. It is not for sales, holidays or investments. Its job is to keep a shock from turning into debt.

How much: a method, not a magic number

No single amount fits everyone. One official source, the US Consumer Financial Protection Bureau, says the right size depends on your situation and suggests basing the goal on unexpected expenses you have faced before (CFPB, accessed 2026-11-07; see Appendix D). Common rules suggest a number of months of essential spending; treat them as a starting range.

Method:

1. Calculate **essential monthly spending**: housing, utilities, food, transport, insurance, minimum debt payments, and any required care costs. Exclude fun, subscriptions you could cancel and savings.
2. Pick a **number of months** from the risk checklist below.
3. Multiply.

4. Build in two stages: a **starter fund** (one small cushion, such as one month of essentials or a figure from your own record of surprise costs) and then the **full fund**.

Choose months by risk:

Situation	Suggested range (months)
Stable job, no dependants, low fixed costs	3
One income for a household or dependants	4 to 6
Variable or freelance income	6 to 9
Specialised job with long search times	6 or more
Health or housing risk, no insurance	add 1 to 3

These ranges are common planning conventions, not rules from a regulator.

Worked examples

Example A (low income). Essential spending 1,200 CU per month, stable job, target 3 months. Target = 1,200 x 3 = 3,600 CU. Starter fund: 1,000 CU. Saving 80 CU per month (from Example A in Chapter 3), the starter fund takes 1,000 / 80 = 12.5, so 13 months, and the full fund takes 3,600 / 80 = 45 months. That is slow. Options: raise the monthly amount using the buffer, direct windfalls (tax refund, gift) into it, or hold the starter fund as the goal for the first year. A slow start is still a start.

Example B (middle income). Essential spending 2,300 CU, one household income, target 6 months. Target = 2,300 x 6 = 13,800 CU. Saving 300 CU per month takes 13,800 / 300 = 46 months with no interest. If the money earned an **assumed** 2 percent a year, added monthly (each month's interest is the balance times 0.02 / 12, rounded

to the cent, then 300 is added), it would take 45 months and reach 14,007.05 CU. Interest helps a little; the contributions do most of the work.

Example C (freelancer). Essential spending 1,550 CU, target 6 to 9 months. 6 months = 9,300 CU; 9 months = 13,950 CU. Saving 180 CU per month (Example C in Chapter 3) takes 9,300 / 180 = 51.67, so 52 months, for the 6 month target. Because that is slow, the freelancer in Chapter 3 also routes surplus months into the fund from the holding account, which can shorten the timeline in good years but is not guaranteed.

Where to keep it

The fund must be safe, easy to reach and boring: an account you can access within a day or two, separate from your everyday account to reduce temptation, and held with an institution whose deposits are protected under your country's rules (check your own regulator; protection schemes and limits differ by country). Do not put it into assets that can fall in value when you need it, which includes most investments.

Rules for using it

Write them down so you can decide in advance:

```
EMERGENCY FUND RULES
Use it for: loss of income, urgent health costs, urgent essential
repairs (home, vehicle needed for work), emergency travel.
Do not use it for: sales, holidays, planned costs, upgrades.
If I use it: I will rebuild it before I increase other goals.
Two-question test: Is it unexpected? Is it necessary?
Both must be yes.
```

Emergency fund planner

```
EMERGENCY FUND PLANNER
Essential monthly spending:                  _______ CU
Months to cover (from risk table):           _______
TARGET (spending x months):                  _______ CU
Starter fund goal:                           _______ CU
Already saved:                               _______ CU
Remaining:                                   _______ CU
Monthly amount I can add:                    _______ CU
Months to starter goal (remaining / monthly): _______
Months to full goal:                         _______
Windfalls to add (tax refund, gifts):        _______
Where the fund is held:                      _______
Review date:                                 _______
```

US example (labelled). The CFPB page opened for this book recommends setting a specific goal, automating transfers (for example splitting direct deposit where an employer allows it), saving windfalls such as tax refunds, and watching your checking balance to avoid overdraft fees. Readers elsewhere should look for the equivalent guidance from their own consumer protection agency.

Common mistakes and fixes

- **Waiting to save until debts are gone.** Save a starter fund first or each surprise goes back on the card.
- **Calling every want an emergency.** Use the two-question test.
- **Parking the fund where it can lose value.** Keep it accessible and stable.
- **Never rebuilding it.** Add "refill" as the first goal after any withdrawal.

CHAPTER 5

Managing debt

Why debt needs its own system

Debt is a contract: you receive money now and pay it back later with a cost. Some debt is a reasonable tool; some is expensive and corrosive. The difference is the interest rate, the terms and whether you can afford the payments. This chapter gives you a way to see all your debts on one page.

Types of debt

Type	Typical features	Watch for
Credit card	Revolving, variable rate, minimum payment	High rates, balance carries forward
Personal loan	Fixed term, fixed payment	Fees, early repayment terms
Student loan	Often special terms, may be income-linked	Rules differ widely by country
Vehicle or mortgage loan	Secured on an asset	Asset can be repossessed if you default
Store or buy-now-pay-later	Short term, may be interest-free at first	Late fees, deferred interest, stacking
Overdraft	Borrowing against your account	Fees and high rates
Informal (family, friends)	Flexible	Relationship risk; agree terms in writing

"Secured" means the lender can take a named asset if you stop paying; "unsecured" means they cannot take it directly, but they can pursue the debt by other legal routes that depend on your country.

Interest rate versus APR

The **interest rate** is the cost of borrowing before other charges, usually stated per year. The **APR** (annual percentage rate) is intended to express the yearly cost of credit, often including some fees, in a standard form so offers can be compared. The exact definition and what must be included are set by law in each country, so read the lender's disclosure and compare the same measure across offers.

Compounding matters. If a card has an **assumed** 18 percent APR with interest added monthly, the monthly rate is 18 / 12 = 1.5%. On a balance of 1,000 CU, the first month's interest is 1,000 x 0.015 = 15.00 CU. Compounded over a year, 1.015^12 - 1 = 19.56% effective annual cost. The gap between 18 and 19.56 is the effect of compounding on the balance. Always compare like with like.

Fixed-payment loan example (assumptions: 10,000 CU, 8% a year, 24 monthly payments). The payment formula is:

```
payment = P x i / (1 - (1 + i)^-n)
P = 10,000   i = 0.08 / 12   n = 24
payment = 452.27 CU (rounded to the cent)
total paid = 452.27 x 24 = 10,854.48 CU
interest paid = 10,854.48 - 10,000 = 854.48 CU
```

Rounding the payment to the cent can leave a few cents of difference on the final payment; lenders adjust the last instalment.

Minimum payments: why they mislead

A minimum payment keeps the account in good standing; it is not a plan to be debt-free. Many cards set the minimum as a percentage of the balance, so the payment shrinks as the balance shrinks and the

payoff stretches out.

Example (assumptions: 2,000 CU balance, 20% APR, minimum is the greater of 25 CU or 3% of the balance, no new spending).

- Month 1 interest: 2,000 x 0.20 / 12 = 33.33 CU. Balance becomes 2,033.33. Minimum: 3% x 2,033.33 = 61.00 CU.
- Paying only the minimum takes **112 months** (9 years and 4 months) and costs **1,798.28 CU** in interest.
- Paying a fixed **100 CU** every month instead takes **25 months** and costs **453.20 CU** in interest.

Compare the two: the fixed payment pays it off about 87 months sooner. Interest is 1,345.08 CU lower (1,798.28 - 453.20). The lesson is general: keep the payment fixed at the level of your first minimum, or higher, even as the minimum falls.

Debt inventory

List every debt on one page. Copy this template.

```
DEBT INVENTORY                        Date: ________
Name | Type | Balance | APR % | Min pay | Due date | Secured?
-----|------|---------|-------|---------|----------|---------
1.
2.
3.
4.
5.
TOTAL balance:            _______
TOTAL minimum payments:   _______
Monthly interest estimate (balance x APR / 12, per debt): _______
Debts that are past due:  _______
```

Fill in the APR from your latest statement, not from memory. If you cannot find a figure, ask the lender in writing.

If you are behind on payments

Missing payments can bring fees, higher rates, damage to your credit record and, for secured debts, loss of the asset. Act early; options shrink as arrears grow.

1. **Prioritise.** Pay essentials and secured debts that protect your housing, transport to work, energy and food before unsecured debt.
2. **Contact the lender before the due date.** Ask what hardship options exist: a payment plan, a temporary reduction, a due-date change. Keep notes of who you spoke to and when.
3. **Use free advice first.** Free, independent debt advice exists in many countries. Do not pay a company that promises to "erase" your debt; check your regulator's guidance on fee-charging firms.
4. **Never ignore letters from a court or collector.** Replies and deadlines often have legal effect.

Free official or charity sources (examples; check your own country).

- **United States (CFPB).** The CFPB's debt collection page tells consumers to ask collectors for the creditor's name, the amount and how to dispute it, says that federal law (the Fair Debt Collection Practices Act) restricts what collectors may say or do, and says you can tell a collector to stop contacting you. It also explains the difference between credit counselors and debt settlement companies. CFPB phone line: (855) 411-2372 (page opened 2026-11-07).
- **United Kingdom (GOV.UK).** National Debtline offers free, confidential and independent advice by phone and webchat in England and Wales: 0808 808 4000, Monday to Friday 9am to 8pm, Saturday 9:30am to 1pm per the GOV.UK page opened 2026-11-07. Check the page for current hours, call costs and

arrangements in Scotland and Northern Ireland. UK readers can also look for the government-backed MoneyHelper service (not opened for this book; verify it yourself).
- **Elsewhere.** Search for your national consumer protection agency or financial regulator and look for the "debt advice" page. Prefer non-profit or government services.

Common mistakes and fixes

- **Paying only minimums on everything.** Choose a strategy (Chapter 6).
- **Taking a new loan to pay a card without comparing total cost.** Compare total interest and fees on both sides, and watch for transfer fees and for rates that jump after an introductory period.
- **Hiding from collectors.** Respond in writing and ask for the debt to be verified.
- **Stopping all saving to pay debt.** Keep the starter emergency fund going.

CHAPTER 6

Debt repayment strategies

Three strategies

All three pay the **minimum on every debt** and send every extra unit to one target debt. They differ only in the target.

- **Avalanche:** target the debt with the highest APR first. This minimises total interest.
- **Snowball:** target the smallest balance first. It clears accounts sooner, which some people find easier to stick to.
- **Hybrid:** clear the smallest debt first for an early win, then switch to avalanche order for the rest.

When a debt is cleared, its minimum payment is added to the extra amount (the "rollover"), so total monthly payment stays constant.

The comparison: same debts, same budget

The numbers below were produced by `verify_math.py`, which simulates each month: interest is added (balance x APR / 12, rounded to the cent), minimums are paid, then the extra goes to the target debt. All rates are **assumptions**.

Debt	Balance (CU)	APR	Minimum (CU)
Card A	3,200	24%	80
Card B	1,100	19%	35

Debt	Balance (CU)	APR	Minimum (CU)
Loan C	5,500	9%	130
Bill D	600	6%	25
Total	**10,400**		**270**

Extra payment: 250 CU a month, so the total monthly debt budget is 520 CU. New borrowing: none. Loan C's minimum is treated as a normal minimum that can be prepaid without penalty (check your own loan's terms).

Orders:

- Avalanche: Card A, Card B, Loan C, Bill D.
- Snowball: Bill D, Card B, Card A, Loan C.
- Hybrid: Bill D, then Card A, Card B, Loan C.

Month 1 working (all strategies share the interest step)

```
Interest: A 3,200 x 0.24 / 12 = 64.00
          B 1,100 x 0.19 / 12 = 17.42 (17.4167 rounded)
          C 5,500 x 0.09 / 12 = 41.25
          D   600 x 0.06 / 12 =  3.00
Total interest month 1 = 125.67
Minimums paid = 270; extra = 250 to the target debt.
Avalanche, end of month 1: A = 3,200 + 64.00 - 80 - 250 =
2,934.00
                           B = 1,100 + 17.42 - 35 = 1,082.42
                           C = 5,500 + 41.25 - 130 = 5,411.25
                           D = 600 + 3.00 - 25 = 578.00
Total owed = 10,005.67
```

Results

Measure	Avalanche	Snowball	Hybrid	Minimums only
Months to debt-free	23	23	23	82
Total interest (CU)	1,288.30	1,431.44	1,388.48	4,917.69

Measure	Avalanche	Snowball	Hybrid	Minimums only
Total paid (CU)	11,688.30	11,831.44	11,788.48	15,317.69
Final-month payment (CU)	248.30	391.44	348.48	22.04

"Minimums only" does not roll over freed-up payments, so it shows the cost of never raising the payment.

Payoff month of each debt:

Debt	Avalanche	Snowball	Hybrid	Minimums only
Bill D	23	3	3	26
Card B	14	6	15	44
Card A	11	15	13	82
Loan C	23	23	23	52

(In the avalanche column Bill D and Loan C both finish in month 23; the last payment clears both.)

Total owed at the end of each month

Month	Avalanche	Snowball	Hybrid
1	10,005.67	10,005.67	10,005.67
2	9,604.96	9,608.71	9,608.71
3	9,197.76	9,209.10	9,209.10
4	8,783.96	8,804.44	8,803.53
5	8,363.43	8,394.06	8,391.06
6	7,936.03	7,977.87	7,971.56
7	7,501.66	7,555.46	7,544.90
8	7,060.18	7,125.76	7,110.94
9	6,611.46	6,688.63	6,669.55

Month	Avalanche	Snowball	Hybrid
10	6,155.39	6,243.95	6,220.61
11	5,691.80	5,791.55	5,763.95
12	5,220.73	5,331.31	5,299.44
13	4,743.24	4,863.06	4,826.93
14	4,259.27	4,386.66	4,347.59
15	3,770.51	3,901.95	3,861.47
16	3,278.14	3,411.21	3,370.43
17	2,782.14	2,916.79	2,875.71
18	2,282.48	2,418.67	2,377.28
19	1,779.13	1,916.81	1,875.11
20	1,272.06	1,411.19	1,369.17
21	761.25	901.77	859.44
22	246.67	388.53	345.89
23	0.00	0.00	0.00

What the comparison shows

- All three finish in the same month here because the total budget is the same and the longest, cheapest debt (Loan C) sets the finish line. That will not always be true.
- Avalanche costs the least interest: 143.14 CU less than snowball (1,431.44 - 1,288.30) and 100.18 CU less than hybrid (1,388.48 - 1,288.30) in this example.
- Snowball removes two accounts by month 6 (Bill D in month 3, Card B in month 6); avalanche removes none until month 11. That is the trade: a lower cost for avalanche, earlier visible progress for snowball.
- The much larger effect than the choice of strategy is **paying more than the minimums**: 3,629.39 CU less interest (4,917.69 - 1,288.30) and 59 fewer months than minimums only.

- The gap between strategies depends on how different the rates are, and how large the balances are. If your rates are similar, the order matters little, so pick the one you will follow.

These results are specific to the assumed debts. They are not a prediction for your situation; rerun the comparison with your own numbers.

Hybrid in practice

A common reading is "one quick win, then maths". Choose the smallest debt only if clearing it frees a meaningful payment or removes a nuisance (a collector's calls, a late-fee risk). Otherwise skip to avalanche.

Other levers (compare carefully)

- **Lower the rate:** ask the lender; compare a balance transfer or consolidation loan on total cost including fees and the rate after any introductory period. Consolidation does not remove debt; it moves it, and works only if you stop adding new balances.
- **Raise the payment:** use a sinking-fund approach, side income or one-time money.
- **Negotiate:** lenders sometimes agree to payment plans if you contact them early.

Debt repayment comparison worksheet

```
DEBT REPAYMENT COMPARISON
Debt budget per month (minimums + extra):   _______ CU
Extra above minimums:                       _______ CU

Strategy   | Order of targets | Debt-free month | Total interest
-----------|------------------|-----------------|---------------
Minimums   |                  |                 |
Avalanche  |                  |                 |
Snowball   |                  |                 |
Hybrid     |                  |                 |
```

```
Chosen strategy: _________  Why: ____________________
Start date: _______  Review date (every 3 months): ______
```

To fill it in without a script, build a spreadsheet with one row per month per debt: opening balance, interest (balance x APR / 12), payment, closing balance, and a rollover cell that adds a cleared debt's minimum to the extra.

Common mistakes and fixes

- **Stopping the minimum on the cleared debt without rolling it over:** move the freed amount to the next target on the same day.
- **Adding new card spending while paying:** freeze new borrowing or the plan never ends.
- **Ignoring fees and 0 percent promotional rates that expire:** note expiry dates in the inventory.
- **Choosing a strategy by what sounds best:** run the comparison once and choose.

CHAPTER 7

Saving systems

Pay yourself first

If you save what is left at month end, it is usually nothing. Pay yourself first means the saving transfer happens **right after income arrives**, before you spend. Automate it: a standing order or scheduled transfer on payday into a separate account. Treat it like a bill.

Setting the amount: start with what you can sustain through a bad month, not an aspirational figure. For example, 10 percent of a 3,600 CU take-home is 360 CU (3,600 x 0.10). Percentages are only a starting point; raise the amount each time your income rises or a debt is cleared (the rollover principle from Chapter 6).

Accounts: separate jobs, separate places

Naming accounts makes the goal visible and stops one pot from covering another. A simple structure:

```
1. Everyday account    - bills and spending
2. Emergency fund      - Chapter 4, rarely touched
3. Sinking funds       - irregular costs (below)
4. Goals               - longer-term targets
5. Investing/retirement- Chapters 9 and 12, long horizon only
```

Some banks allow named sub-accounts; otherwise use a spreadsheet to track each purpose in one account.

Sinking funds: monthly slices of annual costs

A sinking fund saves a fixed amount monthly toward a known, irregular cost.

```
Formula: monthly amount = annual cost / 12
         (or cost / months until it is due)
```

Illustrative annual costs (fictional):

Item	Annual cost (CU)
Car service	420
Annual insurance	540
Gifts and holidays	600
Laptop replacement	900
Medical and dental	360
Total	**2,820**

Monthly amount = 2,820 / 12 = 235 CU. Without the fund, the 900 CU laptop would arrive as a surprise; with it, the money is waiting.

Savings goal planner

```
SAVINGS GOAL PLANNER
Goal:                          _______________
Target amount:                 _______ CU
Date needed:                   _______
Months remaining:              _______
Already saved:                 _______ CU
Monthly amount needed:         (target - saved) / months = _______
CU
Can I afford it? (check against budget): Y / N
If not: lower the target / extend the date / raise income
Where the money is held:       _______________
Progress check dates:          _______
```

Worked examples:

1. Goal 2,400 CU in 12 months, nothing saved: 2,400 / 12 = 200 CU a month.
2. Goal 1,800 CU in 9 months, nothing saved: 1,800 / 9 = 200.00 CU a month.

(If the account earns interest, the monthly amount needed is slightly lower; ignore it for short goals unless the rate is meaningful. Keep money for goals under a few years in stable, accessible places, since investments can fall.)

Order of priorities

A commonly used order, which you may adapt to your situation:

1. Starter emergency fund (a small cushion).
2. Pay down high-interest debt.
3. Grow the emergency fund to its target.
4. Goals and retirement saving, and employer matching contributions where offered (check the terms in your country).
5. Investing beyond retirement accounts.

Debt and saving can run in parallel: a split such as most of the extra to debt and a fixed small amount to the fund is common. Choose the split that stops you from borrowing again.

Making saving stick

- **Remove friction:** automatic transfers, a separate bank for the emergency fund.
- **Add friction to spending:** delete saved card details on shopping sites.
- **Name the goals:** a named account is harder to raid.
- **Reward progress:** track each goal's percentage in the sheet.

Common mistakes and fixes

- **Saving into the everyday account:** spend it by accident. Use a separate account.
- **Target too big, date too short:** the monthly amount fails the budget test; extend the date.
- **No sinking funds:** every annual bill feels like an emergency.

CHAPTER 8

Compound interest

The idea

Interest on interest. Money earns a return, and next period the return itself earns a return. The same mechanism works for debt, so it can build savings or raise a balance owed.

Simple versus compound

- **Simple interest:** earned only on the original amount. 1,000 CU at 5% for 10 years earns 1,000 x 0.05 x 10 = 500 CU, for a total of 1,500 CU.
- **Compound interest:** earned on the growing balance.

Formulas

Lump sum:

```
FV = P x (1 + r/m)^(m x t)
FV = future value    P = amount today
r  = annual rate     m = compounding periods per year
t  = years
```

Regular deposits at the end of each month:

```
FV = D x ((1 + i)^n - 1) / i
D = monthly deposit   i = r / 12   n = months
```

Worked examples (rates are assumptions, not forecasts)

1. **Lump sum, annual compounding.** P = 1,000, r = 5%, m = 1, t = 10. FV = 1,000 x 1.05^10 = **1,628.89 CU**. Compound interest earned is 628.89 versus 500 for simple interest.
2. **Same, compounded monthly.** FV = 1,000 x (1 + 0.05/12)^120 = **1,647.01 CU**. More frequent compounding adds 18.12 CU over ten years.
3. **Longer time.** 1,000 CU at 5% for 30 years = **4,321.94 CU**. Tripling the time more than doubles the final amount, because growth builds on growth.
4. **Small savings goal.** 500 CU at 3% for 5 years, annual compounding = 500 x 1.03^5 = **579.64 CU**.
5. **Regular deposits.** Deposit 200 CU at the end of each month at an assumed 5% a year (i = 0.05/12):

Years	Total deposited (CU)	Future value (CU)
10	24,000	31,056.46
20	48,000	82,206.73
30	72,000	166,451.73

Growth is not a straight line: the 30-year result is far more than three times the 10-year result. This is an illustration at a constant assumed rate. Real investment returns vary year to year and can be negative, so no investment grows in a smooth line like this.

6. **Debt compounding.** A balance of 1,000 CU at an assumed 18% APR, compounded monthly with no payments, grows over 3 years to 1,000 x 1.015^36 = **1,709.14 CU**. Compounding works against borrowers just as it works for savers.

Start early or save more? A comparison

Assume a constant 5% a year, deposits of 200 CU a month, and a goal date at 65 (assumptions).

- **Saver A** deposits 200 a month for 10 years (age 25 to 35), then stops and leaves the balance to grow for another 30 years. Total deposited: 24,000 CU. Balance at 35: 31,056.46 CU. Balance at 65: 31,056.46 x (1 + 0.05/12)^360 = **138,752.30 CU**.
- **Saver B** starts at 35 and deposits 200 a month for 30 years. Total deposited: 72,000 CU. Balance at 65: **166,451.73 CU**.

Saver B ends higher, because they deposited three times as much. What the example shows is the weight of time: Saver A deposited one third as much and reached 83.4% of Saver B's result (138,752.30 / 166,451.73). Both approaches work better together; neither is guaranteed.

The rule of 72 (an approximation)

The rule of 72 estimates doubling time: years to double is roughly 72 divided by the annual percentage rate. It is a **mental shortcut, not an exact answer**. The exact time is ln(2) / ln(1 + r). The table compares them.

Rate	Rule of 72 (years)	Exact (years)	Error of the rule
2%	36.00	35.00	+2.85%
4%	18.00	17.67	+1.85%
6%	12.00	11.90	+0.88%
8%	9.00	9.01	-0.07%
10%	7.20	7.27	-1.00%
12%	6.00	6.12	-1.90%

Error = (rule - exact) / exact. Check: 1,000 CU at 6% for 12 years = 1,000 x 1.06^12 = 2,012.20 CU, slightly more than double, consistent with the exact time being just under 12 years. The rule works best near 8 percent and drifts at very low or very high rates. It also ignores fees, tax and inflation.

The effect of fees

Fees are charged on the balance, so they compound too. Using the 200 CU a month for 30 years example, with an assumed 5% gross return and the annual fee subtracted from the return (a simplification):

Annual fee	Net rate	Value after 30 years (CU)	Lost versus no fee (CU)
0%	5%	166,451.73	0.00
1%	4%	138,809.88	27,641.85
2%	3%	116,547.38	49,904.35

A 1 percentage point fee reduces the end value by 27,641.85 CU, which is 16.6% of the no-fee result (27,641.85 / 166,451.73), though only 1 point of the return was charged. When you compare any investment or account, find the total yearly cost (fund charges, platform fees, advice fees, trading costs) and test it in the formula.

Inflation: nominal versus real

Inflation reduces what money buys. If the assumed return is 5% and inflation is an assumed 2%, the **real** return is (1.05 / 1.02) - 1 = 2.94%, not 3% (subtracting gives an approximation; the exact version divides). Applied to the first example, 1,628.89 CU after 10 years is worth 1,628.89 / 1.02^10 = **1,336.26 CU** in today's buying power. Cash in an account paying less than inflation loses buying power, even though the number rises. Inflation and returns both change from year to year; the figures here are placeholders.

Compound interest calculation template

```
COMPOUND INTEREST WORKSHEET
Lump sum:   FV = P x (1 + r/m)^(m x t)
  P = ____  r = ____  m = ____  t = ____   FV = ____
Monthly deposits:  FV = D x ((1 + i)^n - 1) / i
  D = ____  i = r/12 = ____  n = months = ____   FV = ____
Real value:  FV / (1 + inflation)^t = ____
Fee test: repeat with r reduced by the annual fee = ____
```

Spreadsheet formulas: `=FV(rate/12, years*12, -deposit)` for deposits; `=P*(1+r)^t` for a lump sum; test with a rate of 0 to confirm totals equal deposits.

Common mistakes and fixes

- **Using one rate as a promise.** It is an assumption.
- **Ignoring fees and inflation.** Run both tests.
- **Treating the rule of 72 as exact.** Use the formula for decisions.
- **Forgetting debt compounds.** Pay costly balances first.

CHAPTER 9

Introduction to investing

This chapter is education about concepts. It does not recommend any investment, fund, broker, platform or app, and nothing here is a suggestion to buy or sell anything. Investing involves risk, including the loss of money you put in.

What investing is

Saving keeps money for a use you expect soon, in places meant to hold their value. Investing puts money into assets whose value can rise or fall, such as shares in companies, bonds (loans to governments or companies), property or funds that hold a mix of these, in the hope of a return over a long period. The possibility of a higher return exists because of the possibility of a loss. Anyone who promises a high return with no risk is describing something that does not exist (see Chapter 11).

Return comes from risk, with no guarantee

Higher expected returns usually require accepting larger and less predictable swings. "Expected" does not mean "promised": the actual outcome may be much better or worse, and a long-term average says nothing about the next year.

Volatility and how losses work

Volatility is how much a value moves up and down. Losses are harder to recover than they look:

- A fall of 50% needs a rise of **100%** to get back: 1,000 CU becomes 500, and 500 x 2 = 1,000.
- A fall of 20% followed by a rise of 25% returns to the start: 1,000 x 0.80 = 800; 800 x 1.25 = 1,000.

This is why the time horizon matters (below) and why money needed soon should not be placed where a fall of this size is plausible.

Time horizon

Time horizon is when you will need the money. Longer horizons give more time to recover from falls; shorter ones give less. A common planning idea is that money needed within a few years belongs in stable, accessible places, and that investing is for money you will not need for a long time. A year with a big fall near the date you need the money is a real danger known as sequence risk.

Risk and return in practice: asset classes in general

Type	What it is	Main risks
Cash and deposits	Money in bank accounts	Inflation; limits on deposit protection
Bonds	Loans to governments or companies	Rates, inflation, borrower default
Shares (equities)	Part-ownership of companies	Large swings; company failure
Property	Buildings and land	Illiquid, costs, local market
Funds	Pooled holdings of the above	Fees; inherits the risk of what it holds

Diversification

Diversification means spreading money across many holdings that do not all move together, so one failure does not decide the result. It lowers the risk of a single failure; it does not remove risk. In a broad market fall, many holdings fall at once.

Concentration example (illustrative). You hold 10,000 CU. If it is all in one company and the company's value falls 40%, you lose 4,000 CU. If it is split equally across ten unrelated holdings and one of them falls 40% while the other nine are unchanged, you lose 1,000 x 0.40 = 400 CU. Diversification limited the damage from one failure. It would not have helped if all ten fell.

Funds, "index" and individual holdings: the generic difference

A **fund** pools many people's money to hold many assets. Some funds track a published list of companies (an index) and try to match it; others have a manager who tries to beat a benchmark. Individual securities (one company's shares or bonds) carry company-specific risk that a broad fund spreads out. Funds carry their own risks: fees, the risks of what they hold, and in some structures, the risk of the firm running them. This book does not say which kind to choose. If you are considering a product, read its official fact sheet or disclosure, check that the seller is authorised by your country's regulator, and understand every fee before you commit.

Fees

Fees come in several forms: fund charges, platform or account fees, advice fees, trading costs and taxes. Chapter 8 shows that a 1 percentage point annual charge reduced a 30-year result by 27,641.85 CU on the assumed example. Ask for the total annual cost as a percentage and in currency terms.

Retirement and tax-advantaged accounts, in general terms

Many countries offer accounts designed for long-term or retirement saving that give tax benefits or employer contributions, often with rules about when and how money can be withdrawn. The names, limits, penalties and eligibility are local and change over time. Before choosing, find your country's official explanation (the tax authority or pensions regulator), check whether your employer contributes, and read the withdrawal rules. Do not rely on a summary from a seller of financial products.

Behavioural mistakes

Mistake	What happens	Safeguard
Chasing recent winners	Buy after a rise, sell after a fall	Decide rules before you invest
Panic selling	Locks in losses	Define in advance what you will do in a 30% fall
Overconfidence	Concentrated bets	Limit any single holding
Checking constantly	Stress, impulsive moves	Review on a calendar
Following social media	Hype, hidden promotions	Check source and incentives
Investing emergency money	Forced sales	Fund the cushion first
Ignoring fees	Quiet drag	Calculate total cost

A pre-investing checklist

- [ ] I have a starter emergency fund (Chapter 4).
- [ ] I have a plan for high-interest debt (Chapter 6).
- [ ] The money will not be needed for a long time.

- [ ] I can describe in one sentence what the product holds and how it makes money.
- [ ] I know every fee, in percent and in currency.
- [ ] The seller is authorised by my country's regulator (I checked the official register).
- [ ] I could live with a large fall without selling.
- [ ] I have read the official disclosure document.
- [ ] I have considered advice from a qualified, regulated adviser for my situation.

Common mistakes and fixes

- **Investing before the cushion exists.** Build the cushion first.
- **Treating assumed returns as likely.** Use a range, including zero and negative years.
- **Buying what you cannot explain.** If you cannot explain it simply, wait.

CHAPTER 10

Understanding financial risk

Risk is more than market movement

Risk	What it is	Reduce it by
Income risk	Job or client loss	Emergency fund, skills, several clients
Market risk	Investment values falling	Diversification, time horizon
Inflation risk	Prices rising faster than your money	Not holding long-term money only in cash
Interest-rate risk	Variable payments rising	Fixed rates, margin in budget
Credit and debt risk	Inability to repay	Low fixed costs, avoid stacked debt
Liquidity risk	Cannot turn an asset into cash quickly	Hold accessible savings
Concentration risk	Too much in one place (one employer, asset, country)	Spread
Health and life risk	Illness, injury, death	Insurance, will, plan
Fraud risk	Being deceived	Chapter 11
Longevity risk	Outliving savings	Long-term planning

Risk tolerance versus risk capacity

Tolerance is how much risk you are willing to take; it is emotional. **Capacity** is how much risk you can afford to take; it is financial. They can differ. A person who feels relaxed about large swings but needs the money in two years has high tolerance and low capacity. A cautious person with decades ahead and a secure income has lower tolerance and higher capacity. Plan to the **lower** of the two, and revisit when circumstances change.

```
RISK SELF-CHECK
1. When will I need this money? _____ years
2. If it fell 30 percent, could I avoid selling? Y / N
3. How stable is my income? (stable / variable / uncertain)
4. Would a loss change my ability to pay essentials? Y / N
5. Do I have a cushion that covers 3+ months? Y / N
Any "N" or "uncertain" in questions 2-5 = reduce risk or wait.
```

Insurance basics

Insurance transfers a rare, expensive loss to a company in return for a regular premium. Its purpose is to cover losses you could not absorb, not small ones you could.

Key terms:

- **Premium:** the regular price of cover.
- **Excess or deductible:** the amount you pay yourself before the insurer pays.
- **Limit:** the most the insurer will pay.
- **Exclusions:** situations not covered. Read them.

Example of the trade-off (illustrative). Cover A costs 40 CU a month (480 a year) with a 200 CU excess. Cover B costs 25 CU a month (300 a year) with a 800 CU excess. B saves 180 CU a year (480 - 300), but if you claim, you pay 600 CU more of the cost (800 - 200). B is reasonable only if you hold the higher excess in your emergency fund.

Types people commonly consider (availability and rules vary by country): health, income protection or disability, life (if others depend on your income), home or renters, vehicle, and liability. Only buy cover you understand, from an insurer authorised in your country, and compare terms, not only price.

Concentration

Concentration includes holding mostly your employer's shares, relying on one client for most of your income, or having all savings in one place. Check each year: what single event would cause the largest damage, and what can I do about it?

Building a risk map

```
RISK MAP
Event | How likely? | How bad? | Cushion/insurance? | Action
Job loss               |   |   |   |
Large medical cost     |   |   |   |
Home/vehicle damage    |   |   |   |
Disability             |   |   |   |
Client loss (freelance)|   |   |   |
Market fall            |   |   |   |
```

CHAPTER 11

Avoiding financial scams

Why scams work

Scammers rely on pressure, trust and urgency rather than technology. Anyone can be fooled, particularly when stressed or when a pitch matches a real need. The defence is a fixed set of rules you apply before you pay or share information, regardless of how convincing the message is.

Common patterns

The patterns below are consistent with the guidance on the US Federal Trade Commission's consumer site (pages opened 2026-11-07; see Appendix D). Other countries' agencies publish similar warnings.

Impersonation. The contact claims to be a government agency, bank, utility, tech company or charity. Caller ID and email names can be faked. The FTC's guidance notes that the agency itself will never threaten you or tell you to move money to protect it.

A problem or a prize. You owe money, have a virus, or have won something but must pay a fee first.

Pressure to act now. Threats of arrest, lawsuits or account closure; instructions to stay on the phone.

Unusual payment methods. Gift cards, wire transfers, payment apps and cryptocurrency. Cryptocurrency payments usually cannot be reversed and typically lack the dispute protection that credit cards offer.

Advance-fee scams. You are asked to pay a fee upfront to receive a loan, prize, job or inheritance. The fee is the scam.

Overpayment or fake cheque. Someone sends a cheque for more than owed and asks you to send back the difference before the cheque is found to be invalid.

Romance scams. The scammer builds an online relationship, avoids meeting in person (they claim to live abroad or work on a rig, in the military or for an international organisation), then asks for money for medical bills, travel, visas or crypto investing.

Crypto and "guaranteed return" schemes. The FTC's crypto guidance lists these red flags: guaranteed or large returns (nobody can guarantee profits; "zero risk" or "low risk" claims are false for a volatile asset), free-money offers, faked celebrity endorsements and testimonials, vague explanations, and upfront fees. A common version: a stranger offers to grow your money if you move crypto to their website, which looks real but is fake, and you cannot withdraw.

Pig-butchering-type approaches. A long, friendly online conversation that gradually moves to an "investment" platform. Early small "profits" on a fake dashboard can encourage larger deposits, and withdrawals are then blocked or require more fees. This combines the romance and fake-investment patterns above (the FTC pages describe both components; the term "pig butchering" itself is used in news and regulator coverage that this book did not open directly). The key signal: someone you met online who suggests a specific investment platform.

Phishing. Emails, texts or messages that mimic a real organisation and urge you to click a link or enter a password. Do not use the link; go to the organisation's own site, or call a number you looked up yourself.

Scam-check checklist

Before sending money or information, answer these:

- [ ] Did they contact me first (unsolicited)?
- [ ] Do they claim to be someone official, or someone I have only met online?
- [ ] Are they pressuring me to act now or keep it secret?
- [ ] Do they want gift cards, a transfer, a payment app or crypto?
- [ ] Is there a promise of guaranteed, large or "risk-free" returns?
- [ ] Is there an upfront fee to receive money or a prize?
- [ ] Does the seller appear on my regulator's official register of authorised firms?
- [ ] Have I searched the name plus "scam" or "complaint"?
- [ ] Have I told a trusted person and waited at least one day?
- [ ] Would a legitimate business give me time to decide? (It would.)

One "yes" to the first six is a reason to stop; several is almost certainly a scam.

Protect yourself in advance

- Use unique passwords and two-factor authentication on email and banking.
- Set low limits on payment apps and transfers where possible.
- Agree a family code word for emergency money requests.
- Never share one-time codes or remote-access to your device.
- Keep a list of official contact numbers for your bank, written from its card or site.

If you have been scammed

Act quickly. The steps below follow the FTC's "what to do" guidance (US page opened 2026-11-07). In other countries the same principle applies: contact the payment provider first, then report to your national agency.

1. **Stop contact** with the scammer. Do not send more money to "recover" losses; that is a common second scam.
2. **Contact the payment provider immediately,** using the number on your card or in your official app:
   - bank transfer or payment app: ask for a reversal;
   - wire transfer: report it and ask for a refund;
   - credit or debit card: ask for a refund (chargeback);
   - gift card: contact the card issuer, and keep the card and receipt;
   - cryptocurrency: contact the exchange or ATM operator; recovery is often difficult.
3. **Protect accounts.** If you gave login details, change the password everywhere you used it and turn on two-factor authentication. If someone had access to your computer or phone, update security software, scan, and change passwords.
4. **If you gave identity details,** use your country's identity-theft recovery service. In the US that is IdentityTheft.gov.
5. **Report it.** In the US: ReportFraud.ftc.gov. Elsewhere, report to your national consumer or fraud-reporting agency and to your local police if your country requires it.
6. **Talk to someone.** Shame keeps people from reporting, and scammers rely on it.

Common mistakes and fixes

- **Paying a "recovery" firm.** Anyone who contacts you offering to get your money back is likely running a second scam.
- **Believing screenshots of profits.** They are easy to fake.

- **Using the contact details in the message.** Use details you find independently.

CHAPTER 12

Long-term financial planning

Plan in layers

1. **Goals.** What do you want money to do, and when?
2. **Net worth.** Where do you stand now?
3. **Annual review.** Are you on track?
4. **Life events.** What will change the plan?
5. **Protection.** What if you cannot work, or die?

Setting goals

Use time horizons:

Horizon	Examples	Typical home for the money
Under 2 years	Holiday, laptop, course fees	Stable, accessible savings
2 to 7 years	Deposit for a home, a car, a move	Mostly stable, any investing is a deliberate risk
7+ years	Retirement, long-term wealth	Longer-horizon, with the risks in Chapter 9

Write each goal as a target amount and date (see the goal planner in Chapter 7).

Net worth statement

Net worth is assets minus liabilities. Assets: cash, savings, investments, retirement accounts, valuables you could sell. Liabilities: all debts. Do not count assets you would not sell (a home you live in is counted by some people and excluded by others; choose one approach and keep it).

Illustrative example (fictional):

Assets	CU	Liabilities	CU
Current account	900	Credit card	1,300
Emergency fund	2,400	Student loan	6,800
Retirement account	5,200		
Laptop (resale value)	400		
Total assets	**8,900**	**Total liabilities**	**8,100**

Net worth = 8,900 - 8,100 = **800 CU**. A low number is common early on, especially with student debt; the direction over time matters more than the first figure.

```
NET WORTH STATEMENT                    Date: _________
ASSETS                          LIABILITIES
Cash/current: _______           Credit cards: _______
Emergency fund: _______         Loans: _______
Savings goals: _______          Student loans: _______
Investments: _______            Mortgage: _______
Retirement: _______             Other: _______
Other assets: _______
TOTAL ASSETS: _______           TOTAL LIABILITIES: _______
NET WORTH = assets - liabilities = _______
Change since last statement: _______
```

Update it on a fixed date each year (and a quick check every quarter). Use realistic resale values for property you own.

The annual review

Once a year, ideally on a fixed date:

```
ANNUAL REVIEW
1. Update the net worth statement.
2. Compare to last year: up, flat or down? Why?
3. Savings rate last year: savings / net income = ____ %
4. Which goals were met? Which moved?
5. Debts: balances, rates, any chance to refinance on total cost?
6. Emergency fund: still the right number of months?
7. Insurance and beneficiaries: still right? (Chapter 10)
8. Investments: still suited to risk capacity and horizon? Fees?
9. Income: any change expected next year?
10. Pick three priorities for the next 12 months.
```

Major life events

Event	Consider
New job or pay change	Update budget, raise saving before spending
Moving in with someone or marriage	Shared accounts, who pays what, insurance, wills
Buying a home	Total cost, emergency fund after deposit, rate risk
Child	Cost, insurance, will and guardians
Loss of income	Emergency plan, contact lenders early
Illness or disability	Insurance terms, benefits available in your country
Divorce or bereavement	Professional legal and financial help
Inheritance or windfall	Pause; place in safe holding while you plan
Retirement	Withdrawal strategy, healthcare, longevity

Estate basics at a high level

These rules differ widely by country; take local legal advice.

- **A will** (or the equivalent) says who receives your property. Without one, local law decides.
- **Beneficiary designations** on retirement accounts, insurance and some bank accounts may override a will. Check they are current.
- **Power of attorney** (or equivalent) lets a chosen person manage your affairs if you cannot.
- **A list of accounts and contacts** in a safe place helps your family; do not store passwords in plain text.
- **Guardians** for children are usually named in a will.

Common mistakes and fixes

- **Never doing the net worth number:** it takes 20 minutes a year.
- **Setting goals with no dates.** Add a date and a monthly amount.
- **Ignoring beneficiary forms.** Old designations can send money to the wrong person.
- **Letting lifestyle rise with every raise:** commit part of each raise to savings before it reaches your spending.

Conclusion

A money system is a short list of habits that run without much thought: see your numbers, give each unit a job, protect against shocks, pay off expensive debt in a chosen order, save automatically, and invest cautiously with money you will not need. None of it needs a high income, and none of it is guaranteed to deliver a particular result; it improves your odds of staying in control.

90-day financial action plan

Do this in order. Each week should take about one to two hours.

Month 1: See and stabilise

Week 1: Starting line.

- [ ] Complete the starting statement (Chapter 1).
- [ ] Download two to three months of statements.
- [ ] List all debts in the debt inventory (Chapter 5).

Week 2: Track.

- [ ] Build the income and expense tracker (Chapter 2).
- [ ] Categorise last month's spending; run the spending analysis.
- [ ] Cancel one subscription you have not used in 60 days.

Week 3: Budget.

- [ ] Choose your net income baseline (use the lowest-six method if irregular).

- [ ] Complete the monthly budget worksheet (Chapter 3) with a buffer line.
- [ ] Set up tax set-aside if you are self-employed.

Week 4: Automate and protect.

- [ ] Open a separate account for the emergency fund.
- [ ] Set a payday transfer, even a small one.
- [ ] Fill in the scam checklist's protective steps: password manager, two-factor on email and banking.

Month 2: Cushion and debt

Week 5: Emergency fund.

- [ ] Calculate your target (Chapter 4) and your starter fund.
- [ ] Add one-time money (a refund, a sale of unused goods) if available.

Week 6: Debt strategy.

- [ ] Update APRs from statements.
- [ ] Run the repayment comparison (Chapter 6) and choose a strategy.
- [ ] Contact any lender you are behind with, or a free debt adviser.

Week 7: Start the plan.

- [ ] Pay all minimums on schedule (set reminders).
- [ ] Direct the extra amount to the target debt.
- [ ] Record the starting balances.

Week 8: Review and refine.

- [ ] Do the first monthly review (Appendix A).
- [ ] Compare actual to planned spending; change two lines.

Month 3: Systems and future

Week 9: Sinking funds.

- [ ] List irregular annual costs; compute the monthly amount.
- [ ] Open or label the accounts.

Week 10: Goals.

- [ ] Complete a goal planner for two goals (Chapter 7).
- [ ] Check them against the budget.

Week 11: Learn before investing.

- [ ] Complete the risk self-check and the pre-investing checklist (Chapters 9 and 10).
- [ ] Find your country's official guidance on retirement accounts and employer contributions.
- [ ] Decide whether to seek regulated advice.

Week 12: Protect and plan.

- [ ] Complete the risk map; review insurance and beneficiaries.
- [ ] Write the net worth statement (Chapter 12).

Week 13: Review and set the next 90 days.

- [ ] Do the 90-day review: net worth change, debt balance change, emergency fund progress.
- [ ] Write three priorities for the next quarter.
- [ ] Book the annual review date in your calendar.

What to do when the plan breaks

It will. A missed week, an overspent month or an emergency does not end the plan. Do the next smallest step: update the tracker, rebalance the budget lines, restart the transfer. The aim is a system that survives mistakes, not a record of perfect months.

Appendix A. Monthly financial review checklist

Set a repeating 30-minute calendar event.

```
MONTHLY REVIEW                          Month: ________
INCOME AND SPENDING
- [ ] Recorded all income and spending for the month
- [ ] Compared actual to planned for every line
- [ ] Largest overspend: _______  Reason: _______
- [ ] Adjusted next month's budget for it
DEBT
- [ ] All minimums paid on time
- [ ] Extra payment made to the target debt
- [ ] Balances updated; interest paid this month: _______
- [ ] Checked for new fees or rate changes
SAVING
- [ ] Emergency fund transfer made: _______
- [ ] Sinking funds funded
- [ ] Goal progress updated
PROTECTION
- [ ] Reviewed account activity for anything unfamiliar
- [ ] Any bills or renewals due in the next 60 days: _______
LOOKING AHEAD
- [ ] Planned irregular costs for next month: _______
- [ ] One change for next month: _______
```

Appendix B. Quick formulas and worksheet index

Formulas

```
Cash flow = net income - spending
Savings rate = savings / net income
Monthly interest = balance x APR / 12
Effective annual rate = (1 + APR/12)^12 - 1
Loan payment = P x i / (1 - (1+i)^-n),  i = APR/12
Future value (lump) = P x (1 + r/m)^(m x t)
Future value (deposits) = D x ((1+i)^n - 1) / i
Real value = future value / (1 + inflation)^years
Rule of 72 (approximate) years to double = 72 / rate in percent
Net worth = assets - liabilities
Emergency target = essential monthly spending x months
Sinking fund = annual cost / 12
```

Worksheet index

Worksheet	Location
Starting statement	Chapter 1
Spending analysis	Chapter 2
Income and expense tracker	Chapter 2
Monthly budget worksheet	Chapter 3
Emergency fund planner	Chapter 4
Debt inventory	Chapter 5

Worksheet	Location
Debt repayment comparison	Chapter 6
Savings goal planner	Chapter 7
Compound interest worksheet	Chapter 8
Risk self-check and risk map	Chapter 10
Scam-check checklist	Chapter 11
Net worth and annual review	Chapter 12
90-day plan	Conclusion
Monthly review checklist	Appendix A

Appendix C. Glossary

APR (annual percentage rate): a standardised yearly cost of credit; definition set by local law.

Asset: something you own that has value.

Asset allocation: how money is divided among asset types.

Avalanche method: repaying the highest-rate debt first.

Bond: a loan to a government or company that pays interest.

Budget: a plan assigning income to uses.

Cash flow: money in minus money out over a period.

Compound interest: interest earned on earlier interest.

Credit score or record: a summary of repayment history used by lenders; systems differ by country.

Diversification: spreading money to reduce reliance on one outcome.

Emergency fund: cash held for unplanned essential costs.

Excess or deductible: the part of an insured loss you pay first.

Fund: a pooled investment holding many assets.

Inflation: a general rise in prices over time.

Liability: something you owe.

Liquidity: how quickly an asset can be turned into cash.

Net worth: assets minus liabilities.

Premium: the price of insurance.

Principal: the original amount borrowed or invested.

Real return: return after allowing for inflation.

Rule of 72: an approximation of doubling time.

Secured debt: debt backed by an asset the lender can claim.

Sinking fund: money set aside for a known future cost.

Snowball method: repaying the smallest balance first.

Time horizon: the time until you need the money.

Volatility: how much a value fluctuates.

Appendix D. References and further reading

All pages were opened on 2026-11-07 (accessed 2026-11-07). They support US or UK examples only; other countries have equivalents.

- US Consumer Financial Protection Bureau, "An essential guide to building an emergency fund": https://www.consumerfinance.gov/an-essential-guide-to-building-an-emergency-fund/ (accessed 2026-11-07). Supports the emergency fund approach in Chapter 4.
- US Consumer Financial Protection Bureau, debt collection tools: https://www.consumerfinance.gov/consumer-tools/debt-collection/ (accessed 2026-11-07). Supports the debt collection guidance in Chapter 5.
- GOV.UK, "Get free debt advice" (National Debtline): https://www.gov.uk/national-debtline (accessed 2026-11-07). Supports the UK debt advice example in Chapter 5.
- US Federal Trade Commission, "How to avoid a scam": https://consumer.ftc.gov/articles/how-avoid-scam (accessed 2026-11-07). Supports scam warning signs in Chapter 11.
- US Federal Trade Commission, "What to know about romance scams": https://consumer.ftc.gov/articles/what-know-about-romance-scams (accessed 2026-11-07). Supports romance scam patterns and steps.
- US Federal Trade Commission, "What to do if you were scammed": https://consumer.ftc.gov/articles/what-do-if-you-were-scammed (accessed 2026-11-07). Supports the payment-

method recovery steps.

- US Federal Trade Commission, "What to know about cryptocurrency and scams": https://consumer.ftc.gov/articles/what-know-about-cryptocurrency-and-scams (accessed 2026-11-07). Supports the crypto and guaranteed-return red flags.

Calculations in this book were produced with `verify_math.py` and are logged in `math_check_log.md` in the book folder.